The 34th International Joint Conference on Artificial Intelligence (IJCAI)

Paper ID: 4373

Balance-aware Sequence Sampling Makes Multimodal Learning Better

Zhi-Hao Guan, Qing-Yuan Jiang*, Yang Yang*

Nanjing University of Science and Technology, Nanjing, China

2025.08.30

CONTENTS

Background

Method

Experiments

Conclusion

CONTENTS

Background

Method

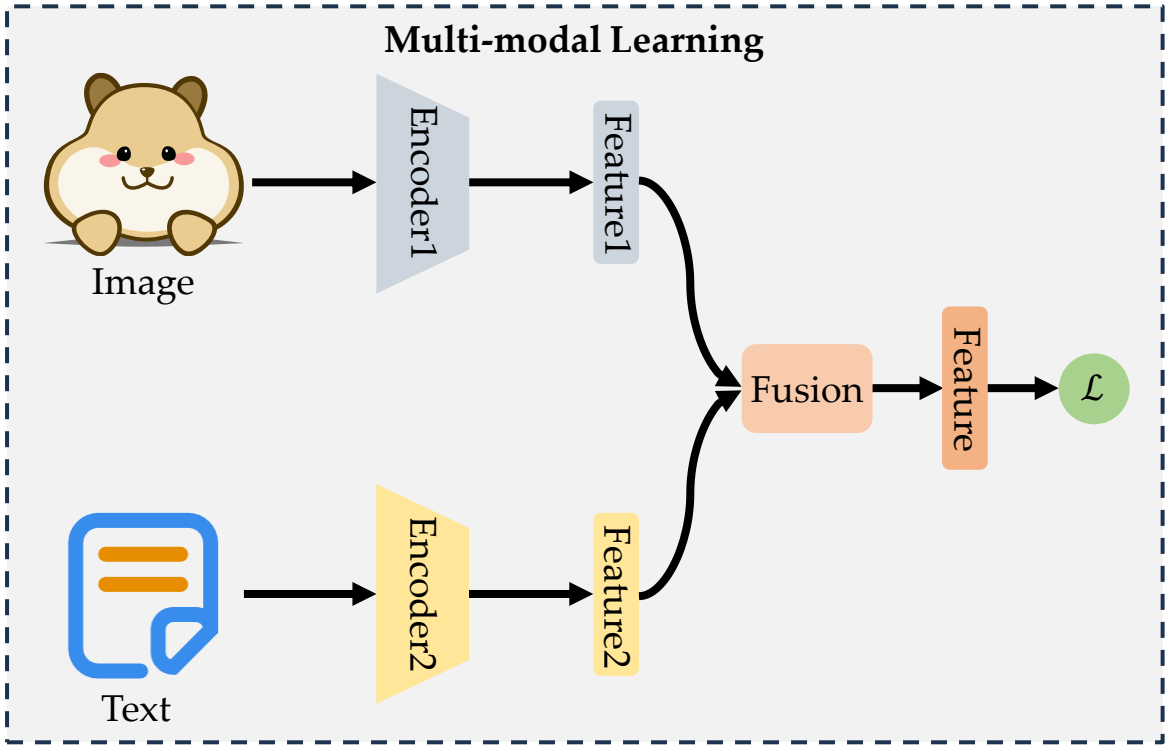
Experiments

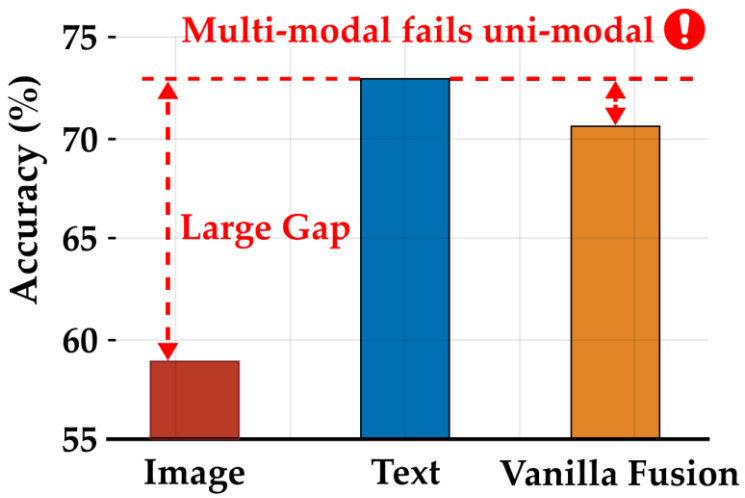
Conclusion

Background

Modality Imbalance

- Due to **modality heterogeneity**, multi-modal learning (MML) is often dominated by stronger modalities, leading to insufficient learning of weaker ones and ultimately suboptimal overall performance.



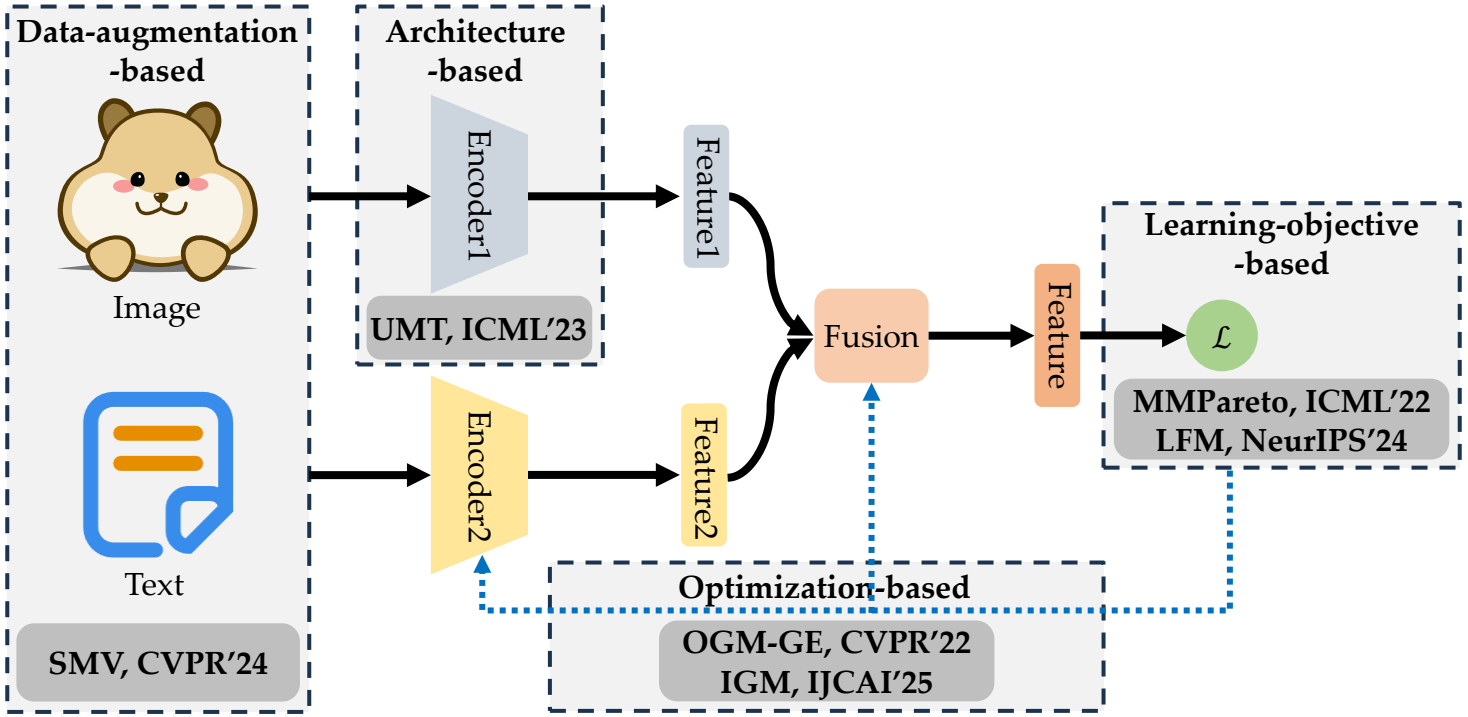


Background

Modality Rebalance Method

- Learning-objective-based: **MMPareto**[ICML'22] , **LFM**[NeurIPS'24]
- Optimization-based: **OGM-GE**[CVPR'22] , **IGM**[IJCAI'25]
- Architecture-based: **UMT**[ICML'23]
- Data-augmentation-based: **SMV**[CVPR'24]

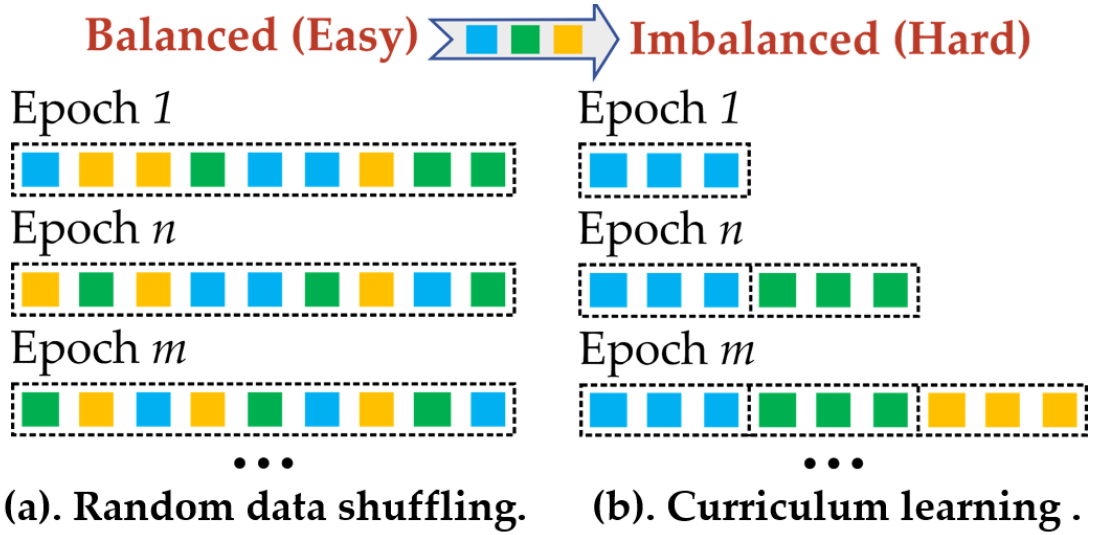
Outstanding Performance!



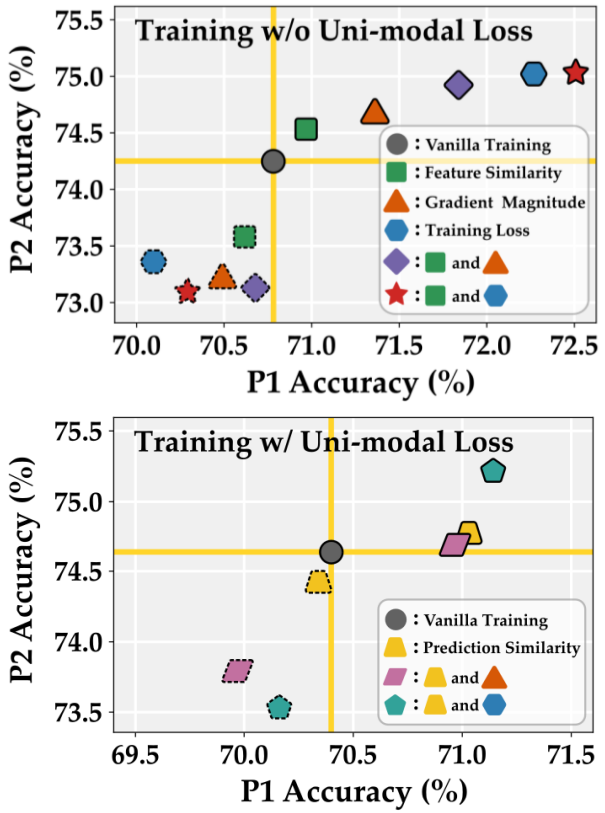
Background

Motivation

- Although existing methods have shown promising results, they generally overlook a key aspect: **MML can be highly sensitive to the training sequence**.



(a). Random data shuffling. (b). Curriculum learning .



CL effectively boosts MML performance, whereas anti-CL degrades it!

CONTENTS

Background

Method

Experiments

Conclusion

Method

Multi-perspective Measurer

- To construct well-structured training sequences that address modality imbalance, we first measure the **balance degree** of a multimodal sample from the following perspectives:

Correlation Criterion

$$\mathtt{sim}(\boldsymbol{x}_i^{(u)}, \boldsymbol{x}_i^{(v)}) = \frac{[\hat{\boldsymbol{y}}_i^{(u)}]^\top \hat{\boldsymbol{y}}_i^{(v)}}{\|\hat{\boldsymbol{y}}_i^{(u)}\|_2 \|\hat{\boldsymbol{y}}_i^{(v)}\|_2}.$$

Information Criterion

$$\ell_{total} = \ell_{multi}^{ce}(\boldsymbol{x}_i, \boldsymbol{y}_i) + \sum_{j \in \{u,v\}} \ell_{uni}^{ce}(\boldsymbol{x}_i^{(j)}, \boldsymbol{y}_i).$$

Balance Score

The balance score of a sample $\boldsymbol{x}_i = \{\boldsymbol{x}_i^{(u)}, \boldsymbol{x}_i^{(v)}\}$ can be formulated as:

$$s(\boldsymbol{x}_i) = \frac{\mathtt{sim}(\boldsymbol{x}_i^{(u)}, \boldsymbol{x}_i^{(v)}) - \mathtt{min}(\mathcal{S})}{\mathtt{max}(\mathcal{S}) - \mathtt{min}(\mathcal{S})} - \frac{\ell_{total}(\boldsymbol{x}_i^{(u)}, \boldsymbol{x}_i^{(v)}, \boldsymbol{y}_i) - \mathtt{min}(\mathcal{L})}{\mathtt{max}(\mathcal{L}) - \mathtt{min}(\mathcal{L})}.$$

Method

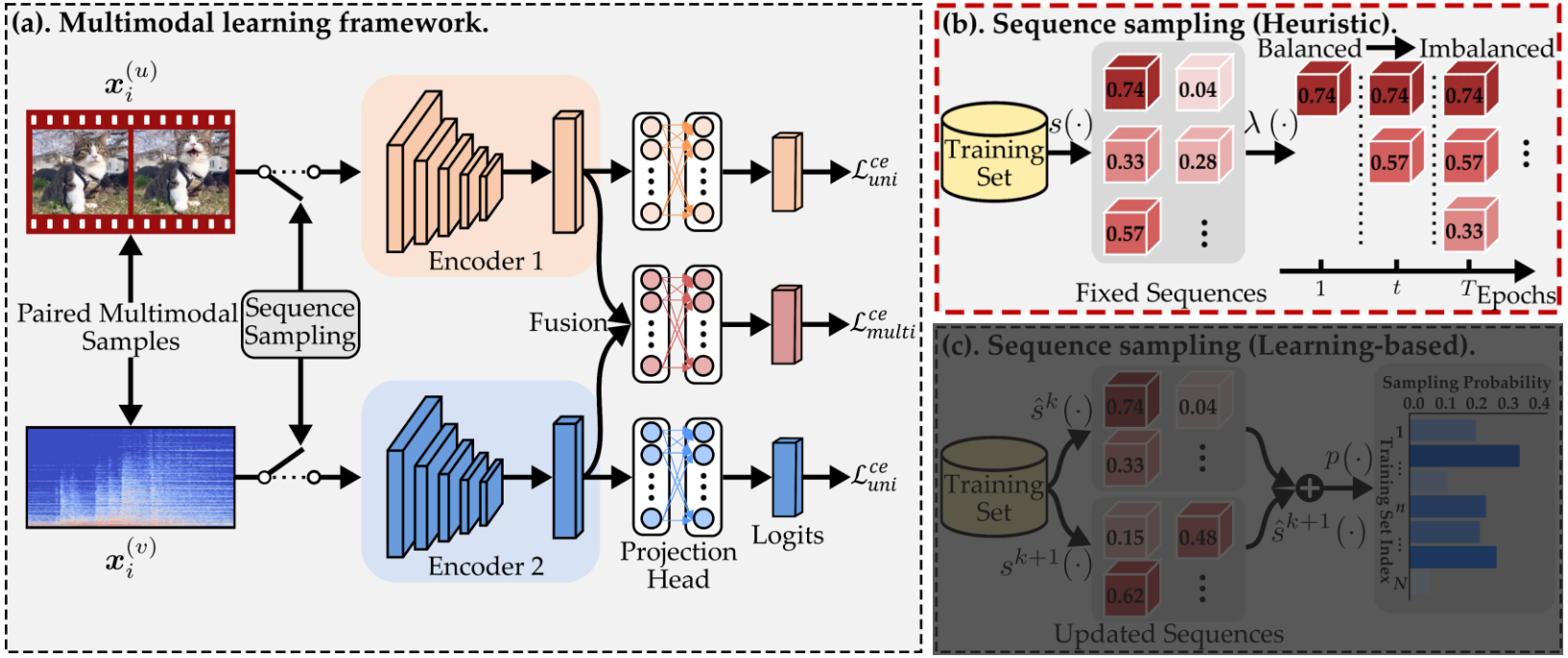
Training Scheduler

- After evaluating the balance score of each sample, we then proceed to control the **presentation order** of training data from balanced to imbalanced samples:

Heuristic Scheduler

$$\lambda(t) = \mathtt{min}\left(1, \sqrt{\frac{1-\lambda_0^2}{T_{grow}} \cdot t + \lambda_0^2}\right).$$

$\lambda(t)$ maps the training epoch t to an interval $\lambda \in (0, 1]$.



At epoch t, the current batch data $\boldsymbol{X}_{batch}$ is randomly sampled from the top λ proportion of the training data in the entire ranked sequence $\boldsymbol{X}_{rank}$:

$$\boldsymbol{X}_{batch}(t) = \mathtt{Sampling}\left(\{\boldsymbol{x}_i | \boldsymbol{x}_i \in \boldsymbol{X}_{rank}, i < \lfloor n \cdot \lambda(t) \rfloor\}\right).$$

Method

Training Scheduler

Leaning-based Scheduler

We update the balance score in a certain epoch E. The $k+1$-th balance score can be denoted as:

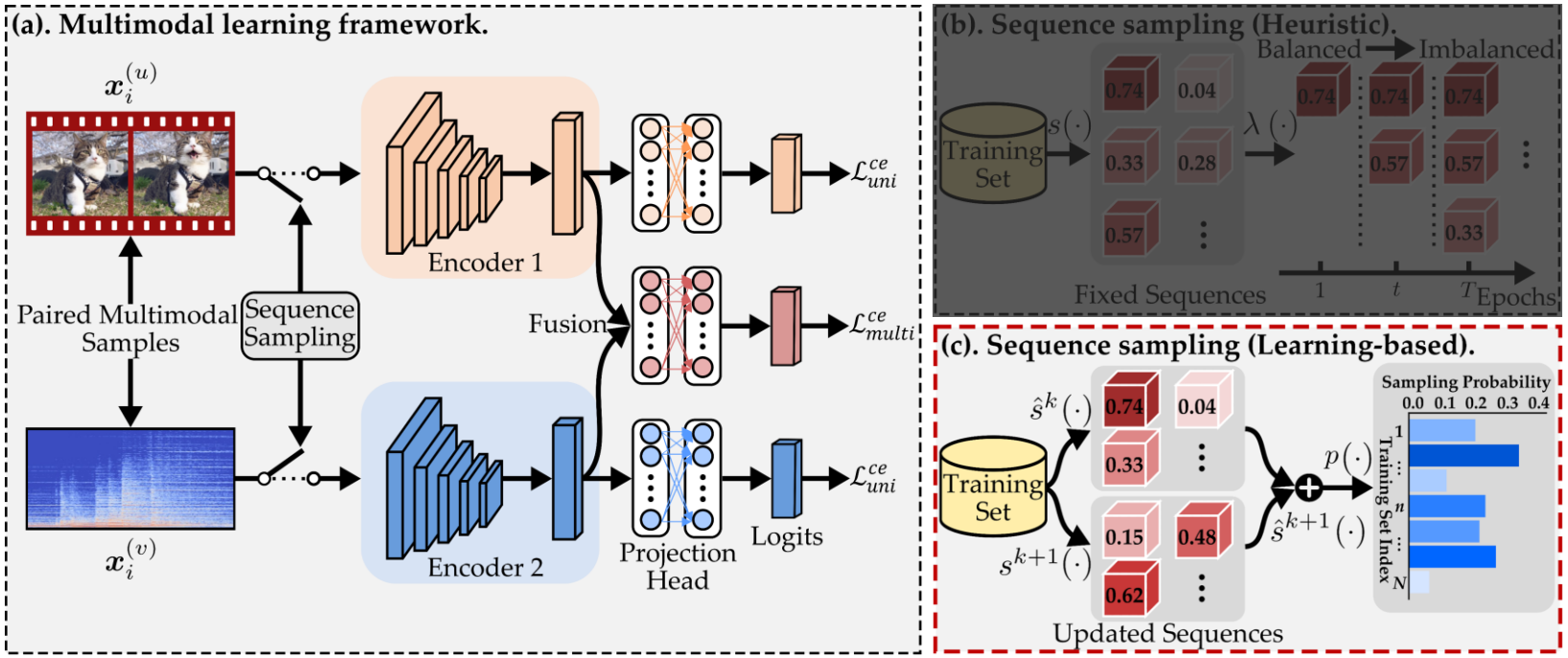
$$\hat{s}^{k+1}(\boldsymbol{x}_i) = \begin{cases} s^{k+1}(\boldsymbol{x}_i), & \text{if } k = 0, \\ (1-\beta)\hat{s}^{k}(\boldsymbol{x}_i) + \beta s^{k+1}(\boldsymbol{x}_i), & \text{otherwise,} \end{cases}$$

where $k = \lfloor t/E \rfloor$, t denotes the t-th epoch.

The sampling probability for each data point $\boldsymbol{x}_i$ can be denoted as:

$$p(\boldsymbol{x}_i) = \frac{e^{\hat{s}^{k+1}(\boldsymbol{x}_i)}}{\sum_{j=1}^{n} e^{\hat{s}^{k+1}(\boldsymbol{x}_j)}}.$$

Finally, $\boldsymbol{x}_i$ is sampled with p to construct the current batch data $\boldsymbol{X}_{batch}$:



$$\boldsymbol{X}_{batch}(t) = \texttt{Sampling}(\{p(\boldsymbol{x}_1), p(\boldsymbol{x}_2), \ldots, p(\boldsymbol{x}_n)\}).$$

CONTENTS

Background

Method

Experiments

Conclusion

Experiments

Classification Results

Table 1: Comparison with SOTA multimodal learning methods. The best performances are highlighted in bold, and the second best is underlined. Higher ACC, MAP, or F1 scores indicate better performance.

Method	CREMA-D		Kinetics-Sounds		Twitter2015		Sarcasm		NVGesture	
	ACC (%)	MAP (%)	ACC (%)	MAP (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)
Audio/Text/RGB	63.17	68.61	54.12	56.69	73.67	68.49	81.36	80.65	78.22	78.33
Video/Image/OF	45.83	58.79	55.62	58.37	58.63	43.33	71.81	70.73	78.63	78.65
Depth	-	-	-	-	-	-	-	-	81.54	81.83
Concat	63.31	68.31	64.55	71.31	70.11	63.86	82.86	82.43	81.33	81.47
Affine	66.26	71.93	64.24	69.31	72.03	59.92	82.47	81.88	82.78	82.81
Channel	66.13	71.75	63.51	68.66	-	-	-	-	81.54	81.57
ML-LSTM	62.94	64.73	63.84	69.02	70.68	65.64	82.05	70.73	83.20	83.30
Sum	63.44	69.08	64.97	71.03	73.12	66.61	82.94	82.47	82.99	83.05
Weight	66.53	73.26	65.33	71.33	72.42	65.16	82.65	82.19	83.42	83.57
ETMC	65.86	71.34	65.67	71.19	73.96	67.39	83.69	83.23	83.61	83.69
MSES	61.56	68.83	64.71	70.63	71.84	66.55	84.18	83.60	81.12	81.47
OGR-GB	64.65	84.54	67.10	71.39	74.35	68.69	83.35	82.71	82.99	83.05
DOMFN	67.34	85.72	66.25	72.44	74.45	68.57	83.56	82.62	-	-
OGM	66.94	71.73	66.06	71.44	74.92	68.74	83.23	82.66	-	-
MSLR	65.46	71.38	65.91	71.96	72.52	64.39	84.23	83.69	82.86	82.92
AGM	67.07	73.58	66.02	72.52	74.83	69.11	84.02	83.44	82.78	82.82
PMR	66.59	70.30	66.56	71.93	74.25	68.60	83.60	82.49	-	-
ReconBoost	74.84	81.24	70.85	74.24	74.42	68.34	84.37	83.17	84.13	86.32
MMPareto	74.87	85.35	70.00	78.50	73.58	67.29	83.48	82.48	83.82	84.24
SMV	78.72	84.17	69.00	74.26	74.28	68.17	84.18	83.68	83.52	83.41
MLA	79.43	85.72	70.04	74.13	73.52	67.13	84.26	83.48	83.40	83.72
AMSS	70.30	76.14	72.25	79.13	75.12	69.23	84.35	83.77	84.64	84.94
BSS-H	80.78	87.86	72.67	78.61	74.73	68.67	84.41	83.86	85.06	85.15
BSS-L	**82.80**	**88.61**	**73.95**	**79.43**	**75.22**	**69.51**	**85.01**	**84.62**	**86.72**	**87.04**

Table 2: Performances on the VGGSound dataset.

Method	ACC (%)	MAP (%)
OGM	48.29	49.78
AGM	47.11	51.98
ReconBoost	50.97	53.87
MMPareto	51.25	54.73
SMV	50.31	53.62
MLA	51.65	54.73
BSS-H	51.61	55.68
BSS-L	**52.80**	**56.61**

Our method achieves **SOTA performance** across various datasets.

Experiments

Further Analysis

- Both "PreSim" and "Loss", can boost classification performance.
- BSS is not sensitive to hyperparameters.
- BSS is robust to the large pre-trained model.

Table 3: Ablation study on the Kinetics-Sounds dataset under the learning-based setting.

Criterion		ACC (%) / MAP (%)		
PreSim	Loss	Audio	Video	Multi
✗	✗	49.37/51.07	54.03/57.48	70.44/76.62
✗	✓	52.11/54.40	54.23/57.91	72.44/79.41
✓	✗	52.38/54.32	**54.93/58.52**	73.25/78.98
✓	✓	**52.73/54.43**	54.74/58.46	**73.95/79.43**

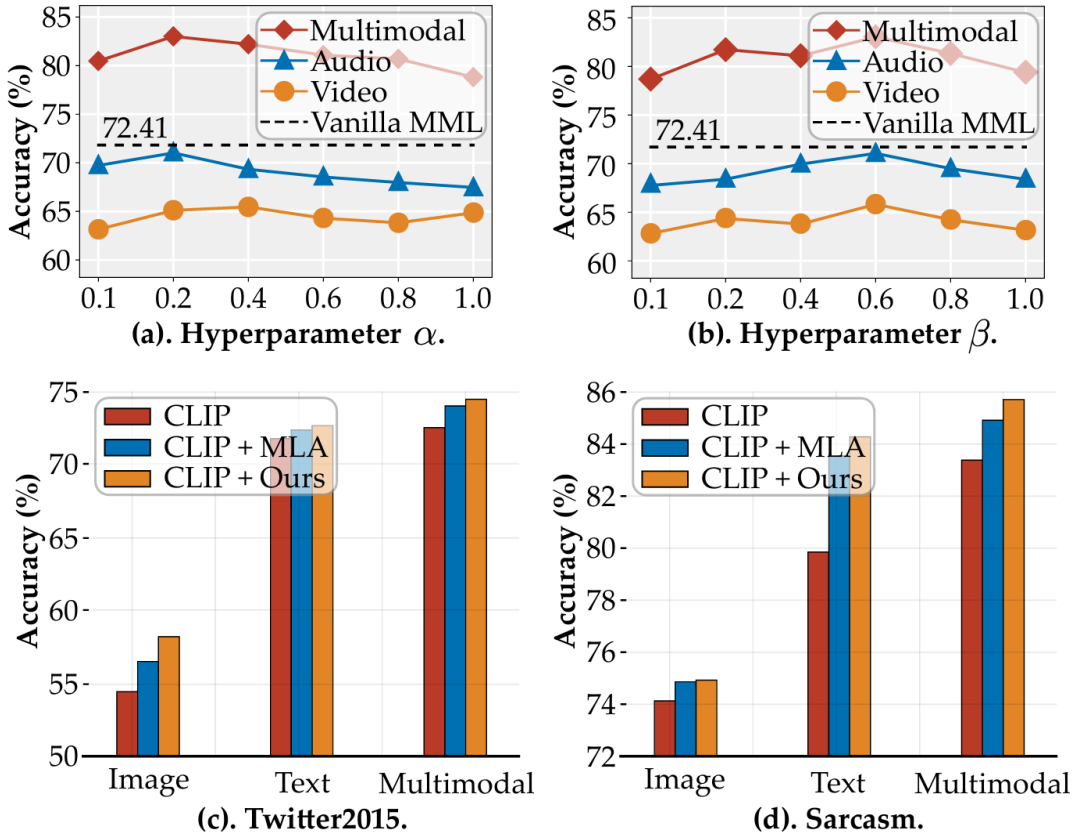


Figure 3: **(a).** and **(b).** Sensitivity to hyperparameters α and β on the CREMA-D dataset. **(c).** and **(d).** Robust performance achieved by using the CLIP pre-trained model as encoders.

CONTENTS

Background

Method

Experiments

Conclusion

Conclusion

Contributions

- We **highlight the critical role of training sequences** in addressing modality imbalance, and show that well-structured sequences can significantly improve MML performance.
- We **define a multi-perspective measurer** to quantify the balance degree of each sample. Based on the resulting balance scores, we then **propose both a heuristic and a learning-based sampling method** to adjust the training sequences.

Future Work

- Expect to extend BSS to other downstream tasks, such as cross-modal retrieval.
- …

Q&A

Thank you for your listening!

KMG Group

WeChat